

From Detection to Association: Learning Discriminative Object Embeddings for Multi-Object Tracking

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Abstract

End-to-end multi-object tracking (MOT) methods have recently achieved remarkable progress by unifying detection and association within a single framework. Despite their strong detection performance, these methods suffer from relatively low association accuracy. Through detailed analysis, we observe that object embeddings produced by the shared DETR architecture display excessively high inter-object similarity, as it emphasizes only category-level discrimination within single frames. In contrast, tracking requires instance-level distinction across frames with spatial and temporal continuity, for which current end-to-end approaches insufficiently optimize object embeddings. To address this, we introduce FDTA (From Detection to Association), an explicit feature refinement framework that enhances object discriminativeness across three complementary perspectives. Specifically, we introduce a Spatial Adapter (SA) to integrate depth-aware cues for spatial continuity, a Temporal Adapter (TA) to aggregate historical information for temporal dependencies, and an Identity Adapter (IA) to leverage quality-aware contrastive learning for instance-level separability. Extensive experiments demonstrate that FDTA achieves state-of-the-art performance on multiple challenging MOT benchmarks, including DanceTrack, SportsMOT, and BFT, highlighting the effectiveness of our proposed discriminative embedding enhancement strategy. The code is available at <https://github.com/Spongebobbobbbbbb/FDTA>.

1. Introduction

Multi-object tracking (MOT) jointly detects multiple objects in video sequences and associates them across frames

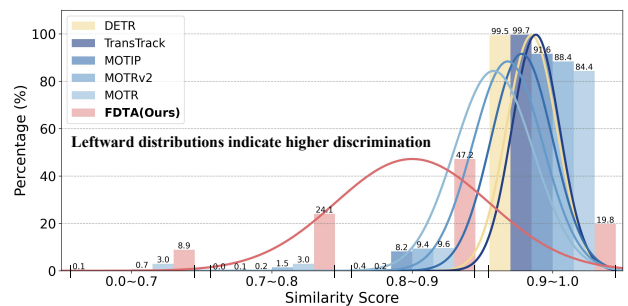


Figure 1. Embedding similarity analysis on DanceTrack. For each frame, we compute pairwise similarities between objects and select the top-3 highest values among all pairs, then construct their distribution across all frames. FDTA produces object embeddings with significantly lower similarity compared to existing methods.

with consistent identities. This technique is critical for a wide range of applications, including autonomous driving [20, 51], action analysis [4, 10, 48], and robotics [13, 31]. Recent *end-to-end* MOT methods [14–16, 34, 46, 52] leverage DETR [7, 26, 61] to generate object embeddings and perform detection and association within an integrated framework, achieving impressive results across multiple challenging benchmarks.

However, end-to-end methods suffer from low association accuracy (AssA~60%), despite achieving high detection performance (DetA> 80%). To investigate the cause, we examine the object embeddings produced by the shared DETR architecture. As shown in Fig. 1, object embeddings from existing methods exhibit severe inter-object similarity, with over 80% of inter-object similarity scores exceeding 0.9. Such high similarity blurs the distinction between different objects, further impairing the association during identification. Notably, the similarity distribution of object embeddings in current end-to-end methods closely mirrors that of the original DETR, which is pretrained purely for detection. The observation suggests an insufficient optimization of discriminative object embeddings for association.

Work done during internship at Shanghai AI Laboratory.

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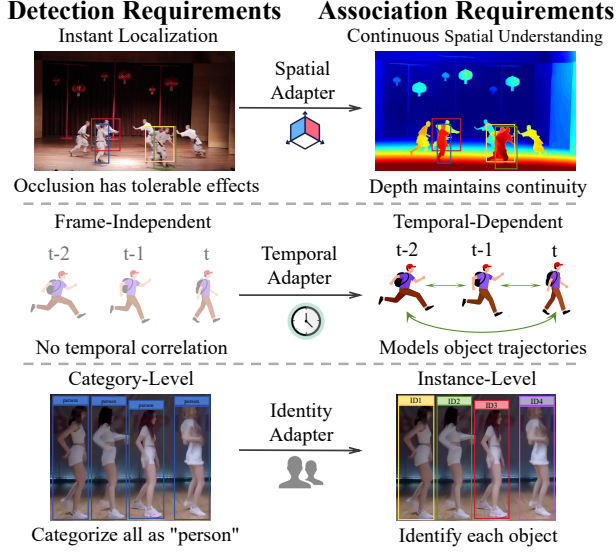


Figure 2. Illustration of the requirements of detection and association. It differs across spatial, temporal, and identity perspectives.

To address this issue, we aim to explicitly refine the object embeddings to enhance their discriminativeness. For effective and targeted refinement, we first clarify the fundamental differences in requirements between detection and association. As shown in Fig. 2, these requirements diverge in three key aspects: (1) **Spatial**: detection relies on instant localization, whereas association demands continuous spatial understanding; (2) **Temporal**: detection treats frames independently, while association requires global context understanding; (3) **Identity**: detection emphasizes category-level discrimination (*person* vs. *car*), while association necessitates instance-level differentiation (*person #1* vs. *person #2*). Due to these disparities, object embeddings learned purely from the detection objective tend to be less discriminative, underscoring the need for additional optimization to make them suitable for association.

Based on the above analysis, we propose **FDTA** (From Detection to Association), an explicit feature refinement framework that learns discriminative embeddings via three complementary modules, each targeting one key aspect.

(1) **Spatial Adapter (SA)**. Depth information complements appearance features extracted for detection, mitigating occlusions by introducing spatial continuity. In SA, we construct a parallel feature extraction branch and introduce depth estimation as an auxiliary task, supervised by pseudo labels from a foundation model [8]. The depth-aware features are then fused with the original object embeddings to enhance spatial discriminativeness.

(2) **Temporal Adapter (TA)**. In TA, temporal dependencies are modeled at the trajectory level, enabling each object embedding to aggregate information across its entire history. Specifically, we design an attention mask tailored for as-

sociation, which accounts for both causality and occlusion, ensuring reliable temporal interactions between frames. As a result, TA enhances the discriminativeness of object embeddings through temporal modeling.

(3) **Identity Adapter (IA)**. To achieve instance-level identification beyond category-level discrimination, IA introduces quality-aware contrastive learning on object embeddings, which naturally aligns with the objective of distinguishing instances. Specifically, it leverages IoU to assess sample quality for guiding identity learning. This contrastive learning encourages embeddings of the same identity to be pulled together while pushing apart those of different identities.

By refinement via three targeted modules, FDTA effectively learns object embeddings suited for tracking, exhibiting stronger discriminative power. As shown in Fig. 1, FDTA significantly reduces inter-object similarity. Consequently, FDTA achieves state-of-the-art performance across multiple benchmarks [11, 38, 58], especially on key tracking metrics HOTA and IDF1.

Our contributions are summarized as follows:

- We identify a common limitation in existing end-to-end MOT methods: the produced object embeddings from DETR exhibit excessively high inter-object similarity, revealing insufficient optimization for tracking.
- We propose FDTA, an explicit feature refinement framework that effectively enhances object discriminativeness through three targeted adapters in spatial, temporal, and identity perspectives.
- Extensive experiments demonstrate that FDTA achieves state-of-the-art performance across various benchmarks, thereby confirming the effectiveness of discriminative object embeddings in improving tracking performance.

2. Related Work

Following prior works [14, 15, 37, 45, 46], we categorize multi-object tracking methods into two main paradigms: *tracking-by-detection* and *end-to-end*.

Tracking-by-Detection. This paradigm decouples detection and association into two sequential stages. Pretrained detectors first generate bounding boxes, which are then linked across frames by association modules [3, 6, 9, 55, 60]. Early methods primarily focus on association algorithms. SORT [3] introduces Kalman filtering, and subsequent works enhance robustness by integrating Re-ID features for appearance matching [43, 54], compensating for camera motion [6], recovering low-confidence detections [55], and refining local matching strategies [35]. More recent approaches, such as TransMOT [9] and GTR [60], employ spatiotemporal transformers for learning-based association. Despite these advances, the decoupled design still inevitably suffers from information loss and error propagation between stages.

End-to-End. To overcome the information loss in the

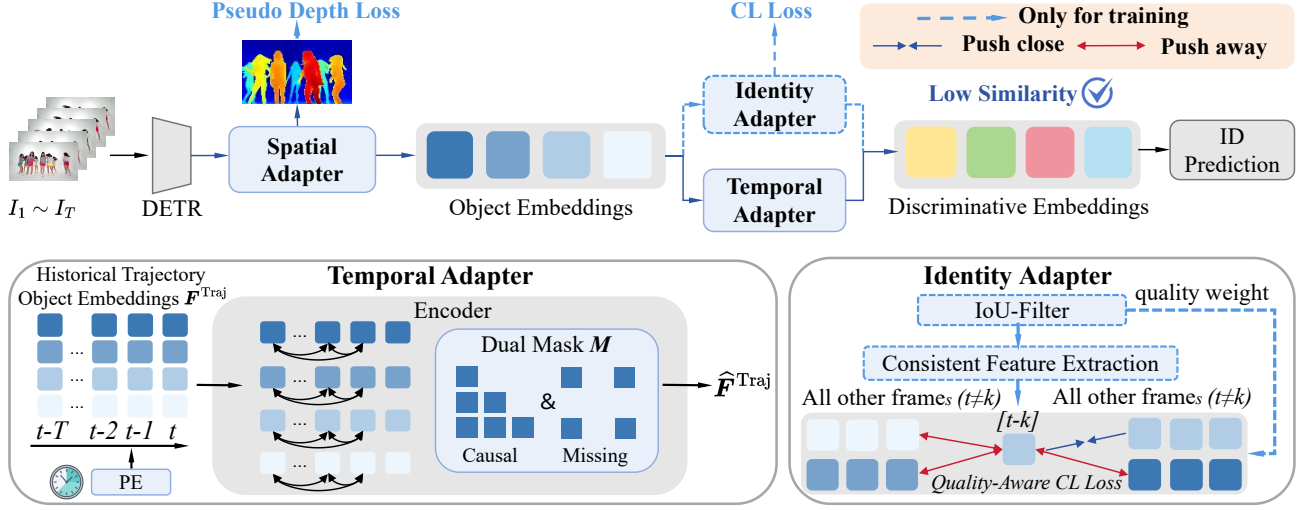


Figure 3. Overview of the FDTA framework. DETR produces object embeddings from input frames. The object embeddings are then refined by three explicit adapters for discriminativeness: Spatial Adapter (SA) integrates 3D geometric cues via depth learning; Temporal Adapter (TA) captures temporal dependencies via trajectory modeling; Identity Adapter (IA) promotes instance-level identification via contrastive learning. Finally, an ID Prediction module performs the object association based on the enhanced embeddings.

decoupled tracking-by-detection paradigm, several methods [14–16, 32, 52, 56] employ DETR [7, 26, 61] to jointly optimize detection and association by generating unified object embeddings, forming an end-to-end paradigm. MOTR [52] uses RNN-like sequential processing to autoregressively propagate track queries across frames. Building on this, MOTRv2 [56] improves query initialization using YOLOX [17], while MeMOT [5] and MeMOTR [15] incorporate memory banks to better aggregate trajectory history. MOTE [14] further enhances robustness by integrating optical flow for occlusion handling. Alternatively, MOTIP [16] enables efficient batch parallel training by reformulating tracking as direct ID classification through learnable ID dictionaries. However, these methods only implicitly optimize object embeddings via the joint detection and tracking losses, lacking explicit constraints for discriminativeness enhancement, which leads to high inter-object similarity and limited tracking performance.

3. Methodology

3.1. Overview

As illustrated in Fig. 3, FDTA builds upon the standard architecture of end-to-end MOT methods [14–16, 52, 56]. To provide context, we first summarize the process used by end-to-end MOT methods. Given a sequence of input frames $\{I_t\}_{t=1}^T$, a shared DETR module [7, 61] is employed to process each frame t independently, generating object embeddings $\{e_i^t\}$ for each object i within the frame. These embeddings are then passed through an ID prediction module to predict object identities, which are used to associate

objects across frames and form object trajectories.

Based on the common patterns of object embeddings observed in Fig. 1, FDTA explicitly incorporates Spatial Adapter (SA), Temporal Adapter (TA), and Identity Adapter (IA) to enhance the discriminativeness of object embeddings from complementary perspectives. Each component is detailed in the following sections.

3.2. Spatial Adapter

Robust association requires continuous spatial understanding across frames to distinguish overlapping objects, while depth provides 3D geometric cues to handle occlusion. Based on this, Spatial Adapter (SA) distills depth knowledge from large-scale pretrained depth estimators, enriching object embeddings with continuous spatial information. **Depth Extraction.** As illustrated in Fig. 4, parallel to the original DETR, SA employs a two-layer convolutional depth extractor to derive dense features, denoted as F_{dense} , from the backbone feature F_V . Additionally, a single-layer convolutional depth head is employed to predict per-pixel depth probabilities $\hat{d}$ over discrete bins through Linear-Increasing Discretization (LID) [53]. We then obtain the depth map $\hat{d}$ by weighting the depth bin values with their predicted probabilities. The details are provided in Sec. B.1. **Depth Distillation.** Benefiting from the generalization capability of pretrained foundation models, we leverage Video Depth Anything [8] to generate offline pseudo depth labels. These continuous labels are discretized into bins $\hat{d}$ using LID for supervision.

Since the tracking task fundamentally prioritizes the object’s region over the background, we introduce a weighted

depth loss to emphasize the quality of foreground depth. Specifically, we utilize the ground truth bounding box to differentiate between foreground and background pixels, and assign a larger penalty weight $w_{i,j}$ to foreground pixels during training. The depth loss is formulated as:

$$\mathcal{L}_{\text{depth}} = \frac{1}{N_{\text{total}}} \sum_{i,j} w_{i,j} \cdot \text{FL}(\mathbf{d}_{i,j}, \bar{\mathbf{d}}_{i,j}), \quad (1)$$

where $\text{FL}(\cdot)$ denotes the Focal Loss [25], $\mathbf{d}_{i,j}$ and $\bar{\mathbf{d}}_{i,j}$ represent the predicted depth probabilities and discretized ground-truth bin at position (i, j) , and $w_{i,j}$ indicates different weights for foreground and background pixels.

Depth Encoding. After obtaining the depth-aware features $\mathbf{F}_{\text{dense}}$ via distillation, we introduce a depth encoding strategy to inject these features into the object embeddings. Specifically, we first utilize a depth encoder, which mirrors the architecture of the standard DETR encoder, to produce refined depth feature $\mathbf{F}_D$. Unlike visual features, the predicted depth map $\hat{\mathbf{d}}$ provides vital positional information for refinement. Accordingly, we compute learnable depth positional embeddings PE_d through linear interpolation:

$$\text{PE}_d = (1 - \delta) \cdot \text{PE}[\lfloor \hat{\mathbf{d}} \rfloor] + \delta \cdot \text{PE}[\lceil \hat{\mathbf{d}} \rceil], \quad \delta = \hat{\mathbf{d}} - \lfloor \hat{\mathbf{d}} \rfloor, \quad (2)$$

where PE denotes the learnable positional embeddings, and $\lceil \cdot \rceil$, $\lfloor \cdot \rfloor$ denote ceiling and floor operations. In the DETR decoder, we additionally add a depth cross-attention layer following standard visual attention layers. It allows the object queries to attend directly to the refined depth features $\mathbf{F}_D$. This yields depth-enriched object embeddings with enhanced spatial discrimination.

Inference Efficiency. Through depth distillation, SA module learns to provide essential depth-aware features $\mathbf{F}_D$, allowing us to deprecate the large foundation model during inference. Notably, this approach represents the first implementation of object embedding enhancement within an end-to-end tracking paradigm. This design ensures both a low computational cost and accurate depth estimation, effectively addressing the limitations of relying on external depth estimators [44, 57] or inaccurate heuristic assumptions based on perspective projections [23, 41]. The detailed computational analysis and depth prediction visualization are provided in Sec. 4.6 and Fig. 5, respectively.

3.3. Temporal Adapter

While the SA module effectively enhances the spatial discriminativeness of object embeddings within individual frames, these frame-independent embeddings inherently lack temporal context across the sequence for association. As a complement, we propose Temporal Adapter (TA) that explicitly captures temporal dependencies across entire trajectories through sequence modeling, enriching embeddings with temporal discriminativeness.

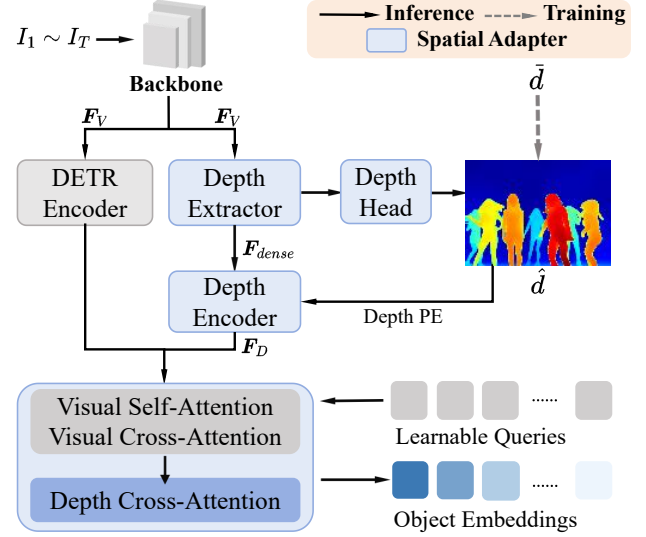


Figure 4. The detailed architecture of Spatial Adapter.

Trajectory Modeling. For online tracking at frame t , we aggregate the historical trajectory $\mathbf{F}_i^{\text{traj}} = \{e_i^{t-T}, \dots, e_i^{t-1}\}$ for each identity i across T previous frames. To capture temporal dependencies, we employ a standard transformer encoder [40] with L layers to process the sequence. However, for standard attention, all tokens attend to each other bidirectionally, which would cause future information leakage and unreliable interactions with [empty] tokens of missing objects. Based on these characteristics of tracking, we design a specialized dual attention mask $\mathbf{M} \in \mathbb{B}^{T \times T}$ that is a union of causal and missing constraints in the binary domain:

$$\mathbf{M}[j, k] = \begin{cases} 1 & \text{if } k > j \text{ or not detected} \\ 0 & \text{otherwise} \end{cases}, \quad (3)$$

where j, k are frame indices. Note that diagonal elements are kept unmasked for numerical stability. The TA module processes the historical trajectory features $\mathbf{F}_i^{\text{traj}}$ using the custom attention mask $\mathbf{M}$ as:

$$\hat{\mathbf{F}}_i^{\text{traj}} = \text{TA}(\mathbf{F}_i^{\text{traj}}, \mathbf{M}), \quad (4)$$

where $\hat{\mathbf{F}}_i^{\text{traj}}$ represents the trajectory embeddings that encode temporal dependencies. Visualization of the temporal interaction weight is provided in Sec. D.5.

3.4. Identity Adapter

Up to this point, SA and TA enhance object embeddings from spatial and temporal perspectives within the scope of tracking objectives. We further focus on exploring explicit optimization objectives directly aligned with the association task. At the instance level, Identity Adapter (IA) leverages quality-aware contrastive learning on the object



Figure 5. Visualization of predicted depth maps on DanceTrack and SportsMOT.

embeddings to learn discriminative features for identification. Specifically, IA aims to pull together embeddings of the same object across different frames while pushing apart embeddings of different objects.

Contrastive Pair Sampling. We construct positive and negative pairs as the foundation for contrastive learning. Specifically, we first aggregate all object embeddings across frames to form the full sample pool. Each object embedding e_i^s at frame s is assigned an identity label ID_i^s using Hungarian matching [21] with the ground-truth bounding boxes. We define positive pair set $\mathcal{P}$ as embeddings that share the same identity, and negative pair set $\mathcal{N}$ as those with different identities. Each pair of embeddings (e_i^s, e_j^k) is categorized as:

$$\begin{aligned} \mathcal{P} &= \{(e_i^s, e_j^k) \mid k \neq s, ID_i^s = ID_j^k\}, \\ \mathcal{N} &= \{(e_i^s, e_j^k) \mid ID_i^s \neq ID_j^k\}. \end{aligned} \quad (5)$$

This sampling strategy allows object embeddings to interact with all others within and across frames. Analogous to the benefits of enlarged sample pools observed in MoCo [19], our design substantially increases the number of negative samples for tracking scenarios.

Quality-Aware Contrastive Learning. Building upon the sampled positive and negative pairs, we apply contrastive learning to effectively enhance object embeddings through two modular designs.

IoU-Filter. In tracking tasks, the samples are generated from predictions, where not all embeddings are equally reliable. Hence, quality control is essential. During identity assignment in sample curation, each object embedding computes its Intersection-over-Union (IoU) score with the corresponding ground truth, which serves as a quality metric. Specifically, we retain only high-quality embeddings with $\text{IoU}_i^t \geq 0.5$ to filter out noisy samples. For each positive pair $(e_i^s, e_j^k) \in \mathcal{P}$, we further assign a weight using the harmonic mean of their IoU scores:

$$w(e_i^s, e_j^k) = \frac{2 \cdot \text{IoU}_i^s \cdot \text{IoU}_j^k}{\text{IoU}_i^s + \text{IoU}_j^k}. \quad (6)$$

Consistent Feature Extraction. Since object embeddings contain frame-variant cues such as motion and pose, directly applying contrastive learning on these embeddings

would interfere with leveraging temporal information. To address this, we employ a 3-layer MLP ϕ to extract identity-consistent features from these embeddings.

Based on the above components, the final quality-aware contrastive learning loss is defined as:

$$\mathcal{L}_{\text{IA}} = \frac{1}{|\mathcal{P}|} \sum_{(e_i^s, e_j^k) \in \mathcal{P}} w(e_i^s, e_j^k) \cdot \mathcal{L}_{\text{InfoNCE}}(e_i^s, e_j^k). \quad (7)$$

Here, $\mathcal{L}_{\text{InfoNCE}}$ denotes the standard InfoNCE loss [39]:

$$\mathcal{L}_{\text{InfoNCE}}(e_i^s, e_j^k) = -\log \frac{\exp(\phi(e_i^s) \cdot \phi(e_j^k)/\tau)}{\sum_{e \in \mathcal{E}} \exp(\phi(e_i^s) \cdot \phi(e)/\tau)}, \quad (8)$$

where the set $\mathcal{E} = \{e \mid (e_i^s, e) \in \mathcal{N}\} \cup \{e_j^k\}$ contains the positive sample e_j^k and all negative samples of e_i^s . τ is the temperature parameter. Note that contrastive learning is applied only during training, thus introducing no additional inference overhead. The IA module is the first to perform quality-aware contrastive learning in an end-to-end tracking framework, clearly distinguishing it from tracking-by-detection methods [12, 36, 49].

3.5. Loss Function

FDTA is trained end-to-end with a combined loss:

$$\mathcal{L} = \mathcal{L}_{\text{det}} + \lambda_{\text{ID}} \mathcal{L}_{\text{ID}} + \lambda_{\text{depth}} \mathcal{L}_{\text{depth}} + \lambda_{\text{IA}} \mathcal{L}_{\text{IA}}, \quad (9)$$

where $\mathcal{L}_{\text{det}}$ comprises standard DETR losses [61]. $\mathcal{L}_{\text{ID}}$ is the cross-entropy loss for ID classification. $\mathcal{L}_{\text{depth}}$ and $\mathcal{L}_{\text{IA}}$ correspond to the depth loss Eq. (1) and quality-aware contrastive loss Eq. (7), respectively. Loss coefficients λ balance each component.

4. Experiment

4.1. Experimental Setup

Datasets. We evaluate FDTA on three challenging benchmarks with similar appearance and complex movements. DanceTrack [38] contains 100 dance videos where dancers wear identical clothing and perform complex synchronized movements. SportsMOT [11] consists of 240 sequences from basketball, volleyball, and soccer scenes, featuring fast-paced motion and frequent occlusions in competitive sports. BFT [58] includes 106 bird flock clips from the BBC documentary Earthflight, featuring complex aerial dynamics, varying formations, and large-scale flocking behavior.

Evaluation Metrics. The evaluation follows standard MOT protocols, using comprehensive metrics including Higher Order Tracking Accuracy (HOTA) [29], ID F1 Score (IDF1) [33], Association Accuracy (AssA), Multi-Object Tracking Accuracy (MOTA) [2], and Detection Accuracy (DetA). Among these, HOTA, IDF1, and AssA primarily assess tracking performance.

Table 1. Performance comparison with state-of-the-art methods on DanceTrack test set. The best result for each metric is shown in **bold**. * indicates methods using extra training data.

Methods	HOTA↑	IDF1↑	AssA↑	MOTA↑	DetA↑
<i>Tracking-by-Detection:</i>					
SORT [3] (ICIP2016)	47.9	50.8	31.2	91.8	72.0
DeepSORT [43] (ICIP2017)	45.6	47.9	29.7	87.8	71.0
CenterTrack [59] (ECCV2020)	41.8	35.7	22.6	86.8	78.1
FairMOT [54] (IJCV2021)	39.7	40.8	23.8	82.2	66.7
QDTrack [12] (CVPR2021)	45.7	44.8	29.2	83.0	72.1
ByteTrack [55] (ECCV2022)	47.3	52.5	31.4	89.5	71.6
OC-SORT [6] (CVPR2023)	55.1	54.2	38.0	89.4	80.3
Hybrid-SORT [47] (AAAI2024)	65.7	67.4	—	91.8	—
SparseTrack [27] (TCSVT2025)	55.7	58.1	39.3	91.3	79.2
DiffMOT [30] (CVPR2025)	62.3	63.0	47.2	92.8	82.5
TrackTrack [35] (CVPR2025)	66.5	67.8	52.9	93.6	—
<i>End-to-End:</i>					
TransTrack [37] (arXiv2020)	45.5	45.2	27.5	88.4	75.9
MOTR [52] (ECCV2022)	54.2	51.5	40.2	79.7	73.5
MeMOTR [15] (ICCV2023)	63.4	65.5	52.3	85.4	77.0
CO-MOT [46] (CVPR2023)	69.4	71.9	58.9	91.2	82.1
MOTRV2 [56] (CVPR2023)	69.9	71.7	59.0	91.9	83.0
MOTIP [16] (CVPR2025)	67.5	72.2	57.6	90.3	79.4
SambaMOTR [34] (ICLR2025)	67.2	70.5	57.5	88.1	78.8
Ours	71.7	77.2	63.5	91.3	81.0
<i>with extra data:</i>					
MOTRV2* [56] (CVPR2023)	73.4	76.0	64.4	92.1	83.7
MOTRV3* [50] (arXiv2023)	70.4	72.3	59.3	92.9	83.8
MOTIP* [16] (CVPR2025)	71.4	76.3	62.8	91.6	81.3
Ours*	74.4	80.0	67.0	92.2	82.7

4.2. Implementation Details

FDTA is built upon Deformable DETR [61] with ResNet-50 [18] backbone, initialized with COCO [24] pretrained weights. We train with video sequences of $T = 30$ frames per batch. For SA, the foreground weighting factor w is set to 7. For TA, the encoder consists of $L = 6$ layers. For IA, we set the temperature parameter $\tau = 0.1$. We train FDTA for 11 epochs with batch size 4 on 4 NVIDIA H200 GPUs. Following prior work [16, 34, 52, 56], we apply standard data augmentation, including random resize, crop, and flip, along with trajectory occlusion and identity switching. We use AdamW optimizer [28] with initial learning rate 1×10^{-4} and weight decay 5×10^{-4} . The loss weights are set as $\lambda_{\text{depth}} = 1.0$, $\lambda_{\text{ID}} = 1.0$, and $\lambda_{\text{IA}} = 1.0$ based on validation performance.

4.3. State-of-the-Art Comparison

DanceTrack. As shown in Tab. 1, FDTA achieves state-of-the-art performance with 71.7% HOTA, 77.2% IDF1, and 63.5% AssA, outperforming the previous best method MOTRV2 by +1.8%, +5.5%, and +4.5% respectively. Training with additional validation data further improves performance to 74.4% HOTA and 80.0% IDF1 on the test set. These gains in association metrics are notable, considering DanceTrack’s extreme inter-object similarity from identical

Table 2. Performance comparison with state-of-the-art methods on SportsMOT test set. The best result for each metric is shown in **bold**.

Methods	HOTA↑	IDF1↑	AssA↑	MOTA↑	DetA↑
<i>Tracking-by-Detection:</i>					
CenterTrack [59] (ECCV2020)	62.7	60.0	48.0	90.8	82.1
FairMOT [54] (IJCV2021)	49.3	53.5	34.7	86.4	70.2
QDTrack [12] (CVPR2021)	60.4	62.3	47.2	90.1	77.5
GTR [60] (CVPR2022)	54.5	55.8	45.9	67.9	64.8
ByteTrack [55] (ECCV2022)	62.8	69.8	51.2	94.1	77.1
BoT-SORT [1] (arXiv2022)	68.7	70.0	55.9	94.5	84.4
OC-SORT [6] (CVPR2023)	68.1	68.0	54.8	93.4	84.8
DiffMOT [30] (CVPR2025)	72.1	72.8	60.5	94.5	86.0
<i>End-to-End:</i>					
TransTrack [37] (arXiv2020)	68.9	71.5	57.5	92.6	82.7
MeMOTR [15] (ICCV2023)	68.8	69.9	57.8	90.2	82.0
MOTIP [16] (CVPR2025)	71.9	75.0	62.0	92.9	83.4
SambaMOTR [34] (ICLR2025)	69.8	71.9	59.4	90.3	82.2
Ours	74.2	78.5	65.5	93.0	84.1

Table 3. Performance comparison with state-of-the-art methods on BFT test set. The best result for each metric is shown in **bold**.

Methods	HOTA↑	IDF1↑	AssA↑	MOTA↑	DetA↑
<i>Tracking-by-Detection:</i>					
SORT [3] (ICIP2016)	61.2	77.2	62.3	75.5	60.6
JDE [42] (ECCV2020)	30.7	37.4	23.4	35.4	40.9
CenterTrack [59] (ECCV2020)	65.0	61.0	54.0	60.2	58.5
FairMOT [54] (IJCV2021)	40.2	41.8	28.2	56.0	53.3
CSTrack [22] (TIP2022)	33.2	34.5	23.7	46.7	47.0
ByteTrack [55] (ECCV2022)	62.5	82.3	64.1	77.2	61.2
OC-SORT [6] (CVPR2023)	66.8	79.3	68.7	77.1	65.4
<i>End-to-End:</i>					
TransCenter [45] (arXiv2021)	60.0	72.4	61.1	74.1	66.0
TransTrack [37] (arXiv2020)	62.1	71.4	60.3	71.4	64.2
TrackFormer [32] (CVPR2022)	63.3	72.4	61.1	74.1	66.0
SambaMOTR [34] (ICLR2025)	69.6	81.9	73.6	72.0	66.0
MOTIP [16] (CVPR2025)	70.5	82.1	71.8	77.1	69.6
Ours	72.2	84.2	74.5	78.2	70.1

clothing and synchronized movements, which highlights the effectiveness of our approach. Qualitative visual comparisons are provided in Fig. 6.

SportsMOT. As shown in Tab. 2, FDTA achieves state-of-the-art performance. The improvements on association metrics demonstrate the effectiveness of FDTA in handling fast-paced motion and frequent occlusions in competitive sports.

BFT. To further validate on non-human scenarios, we evaluate on BFT featuring large flocks with extreme density and rapid formation changes. As shown in Tab. 3, FDTA achieves state-of-the-art performance across all metrics. The consistent improvements across diverse scenarios, from human crowds to competitive sports to bird flocks, demonstrate that our approach is broadly applicable to a wide range of tracking scenarios.

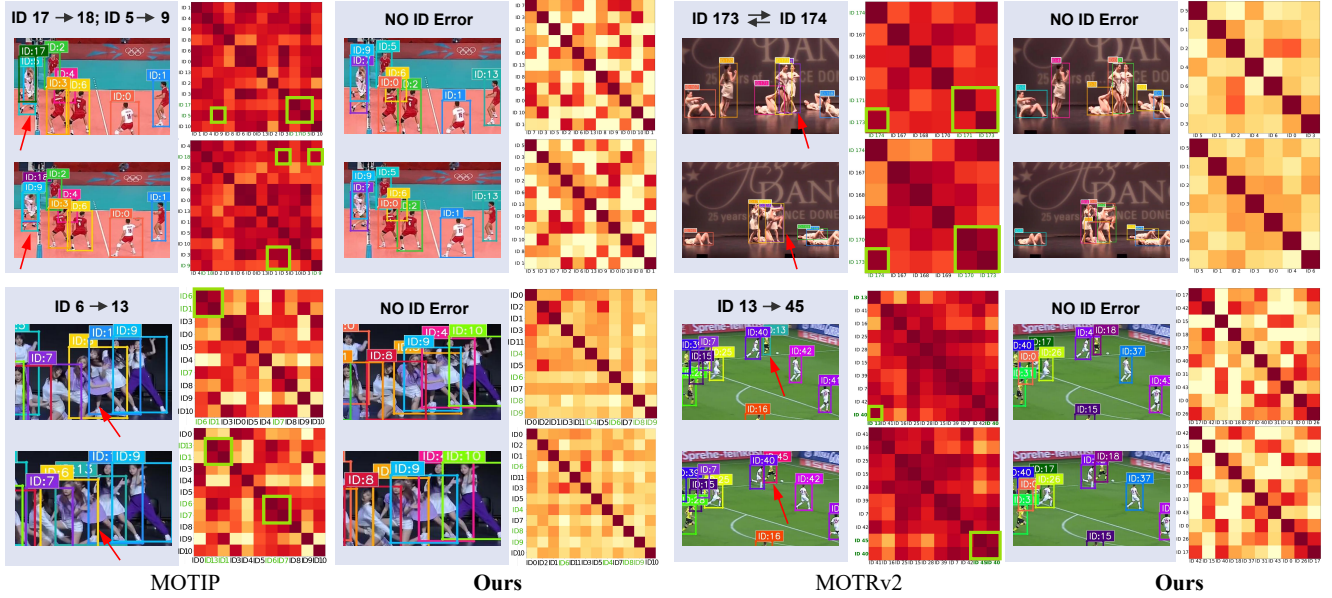


Figure 6. Tracking results visualization and inter-object embedding similarity matrix comparison on DanceTrack and SportsMOT. Darker colors in the similarity matrix indicate higher similarity. Green boxes in the similarity matrix highlight high-similarity regions. Incorrect IDs are marked in the tracking results.

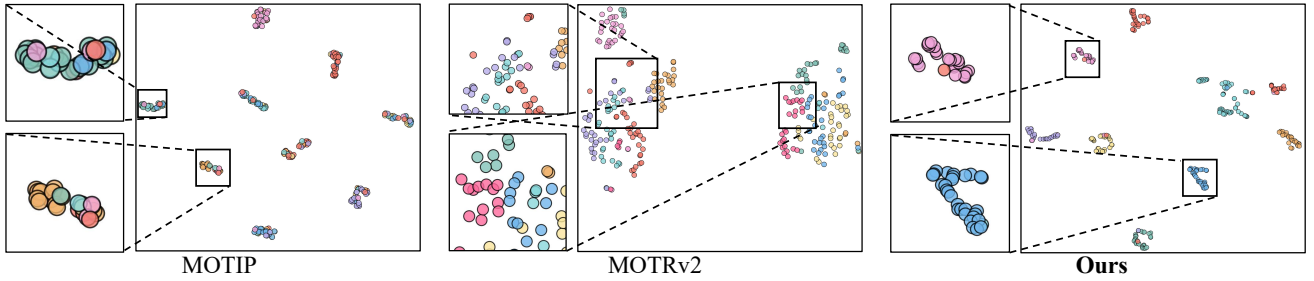


Figure 7. t-SNE visualization of object embeddings on DanceTrack sequence dancetrack0015. Each color represents a tracked object.

4.4. Qualitative Analysis

Tracking Results and Embedding Visualization. We visualize tracking results and inter-object embedding similarity matrix of two strong baselines, MOTIP [16] and MOTRv2 [56], on DanceTrack and SportsMOT. As shown in Fig. 6, both baseline methods exhibit ID errors (red arrows). By examining their embedding similarity matrix, we find these ID errors consistently correlate with high inter-object similarity in the embedding space (green boxes). This validates our key insight that high feature similarity between different objects degrades tracking performance. In contrast, our method produces more discriminative embeddings with reduced inter-object similarity, achieving stable tracking with fewer ID errors. More visualizations are provided in Sec. D.5.

Embedding Space Visualization. To further analyze embedding discriminability, we apply t-SNE to visualize object embeddings across a tracking sequence, where each

color represents one tracked object. Unlike single-frame similarity matrix, t-SNE shows temporal embedding patterns across multiple frames. As shown in Fig. 7, existing methods exhibit two critical issues: (1) overlapping clusters where different objects lack clear boundaries, and (2) mixed clusters containing embeddings from multiple objects, while our method groups same-identity embeddings tightly together and separates different identities distinctly.

4.5. Ablation Studies

We conduct ablation studies on the DanceTrack test set to validate our approach, including module-level ablations and detailed design choices within each adapter.

Complementary Effects of Three Adapters. To evaluate each adapter’s contribution, we progressively add them as shown in Tab. 4. Rows 2-4 show that each adapter individually improves performance. Rows 5-7 show that combining any two adapters yields further improvements. Our complete FDTA (Row 8) achieves 71.7% HOTA and 77.2%

Table 4. Ablation of the proposed adapters. SA: Spatial Adapter; TA: Temporal Adapter; IA: Identity Adapter.

	SA	TA	IA	HOTA↑	IDF1↑	AssA↑	MOTA↑	DetA↑
1	×	×	×	69.4	74.5	60.2	90.6	80.0
2	✓	×	×	70.2	74.8	61.2	90.9	80.7
3	×	✓	×	70.4	75.7	61.3	91.3	81.1
4	×	×	✓	70.1	74.8	60.7	91.2	81.2
5	✓	✓	×	70.8	76.8	61.9	91.2	81.2
6	✓	×	✓	70.6	75.7	62.0	91.1	80.8
7	×	✓	✓	71.0	76.5	62.2	91.3	81.2
8	✓	✓	✓	71.7	77.2	63.5	91.3	81.0

Table 5. Ablation of Depth PE in Spatial Adapter.

Depth Encoding	HOTA↑	IDF1↑	AssA↑	MOTA↑	DetA↑
w/o Depth PE	71.3	76.9	63.1	91.2	81.0
w/ Depth PE	71.7	77.2	63.5	91.3	81.0

Table 6. Ablation of dual attention mask in Temporal Adapter.

Missing Object Handling	HOTA↑	IDF1↑	AssA↑	MOTA↑	DetA↑
w/o TA	69.4	74.5	60.2	90.6	80.0
w/ Zero Vector	68.1	71.6	57.5	91.1	80.9
w/ Missing Mask	70.4	75.7	61.3	91.3	81.1

IDF1, outperforming all other combinations and showing that enhancing object embeddings from spatial, temporal, and identity perspectives is effective and complementary.

Spatial Adapter Design. We examine the effectiveness of Depth Positional Encoding by enabling and disabling this module. As shown in Tab. 5, when directly using raw depth features without Depth PE, performance drops by 0.4% HOTA and 0.3% IDF1, demonstrating that encoding depth as learnable positional embeddings further enhances spatial understanding. Additional ablation studies on foreground weighting and depth encoding strategies are provided in Sec. D.3.

Temporal Adapter Design. TA models temporal dependencies through trajectories with a causal mask to prevent information leakage. We examine the impact of handling missing objects in Table 6. Replacing missing objects with zero vectors (Row 2) degrades performance by 1.3% HOTA and 2.9% IDF1, performing worse than without TA (Row 1). Our missing mask that distinguishes present and missing objects (Row 3) improves performance by 1.0% HOTA and 1.2% IDF1, demonstrating that proper handling of missing objects is crucial for trajectory modeling. Further ablations demonstrating that TA leverages temporal information are provided in Sec. D.4.

Identity Adapter Design. We evaluate three essential components of IA in Tab. 7: Contrastive Learning (CL), Consistent Feature Extractor (CFE), and IoU-Filter (IF). Directly applying CL on raw embeddings (Row 2) degrades performance by 1.0% HOTA and 1.6% IDF1 compared to Row 1. This occurs because raw embeddings contain frame-variant cues that interfere with learning stable identity features. Introducing CFE (Row 3) projects embeddings into an identity-specific space, improving performance by 0.5%

Table 7. Ablation study on Identity Adapter. CL: Contrastive Learning; CFE: Consistent Feature Extractor; IF: IoU-Filter.

CL	CFE	IF	HOTA↑	IDF1↑	AssA↑	MOTA↑	DetA↑
×	×	×	69.4	74.5	60.2	90.6	80.0
✓	×	×	68.4	72.9	58.3	90.7	80.6
✓	✓	×	69.9	74.5	60.5	91.0	80.8
✓	✓	✓	70.1	74.8	60.7	91.2	81.2

Table 8. Computational breakdown on 1920×1080 resolution. Gray rows indicate our proposed adapters.

Component	Time (ms)	% of Total
DETR	62.71	83.9%
SA	1.02	1.4%
TA	1.99	2.7%
ID Prediction	6.67	8.9%
Other Components	2.37	3.2%
Total	74.76	100%

HOTA over Row 1 and recovering the degradation from Row 2 by 1.5% HOTA and 1.6% IDF1. This demonstrates that extracting stable identity features is essential for effective contrastive learning. Adding IF (Row 4) further improves performance by 0.2% HOTA and 0.3% IDF1 over Row 3, confirming that IoU-based sample weighting effectively enhances the contrastive objective by prioritizing reliable positive pairs.

4.6. Computational Analysis

To assess the computational overhead introduced by our proposed components, Table 8 provides a detailed breakdown of inference time on the DanceTrack at 1920×1080 resolution using an H200 GPU. SA and TA (highlighted in gray) introduce minimal overhead of only 1.4% and 2.7%, respectively. Notably, IA is absent from this table as it functions solely during training, incurring zero inference cost. DETR dominates at 83.9%, inherent to the base architecture. Overall, FDτα maintains comparable speed at 13.4 FPS (74.76 ms per frame), improving tracking performance without compromising speed.

5. Conclusion

We identify that object embeddings generated by shared DETR exhibit severe inter-object similarity, which inadequately serves tracking requirements. To address this, we propose FDτα to explicitly enhance object embeddings through three complementary perspectives: Spatial Adapter for 3D geometric understanding, Temporal Adapter for temporal modeling, and Identity Adapter for instance-level separation. Extensive experiments demonstrate that FDτα achieves state-of-the-art performance across DanceTrack, SportsMOT, and BFT, while maintaining efficient inference speed. Future work will leverage foundation models such as video generation and world models to synthesize challenging corner cases for enhancing tracking robustness in extreme scenarios.

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